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USED IN LIEU OF
MIL-T-36487 (DSA-DM)
25 October 1966

MILITARY SPECIFICATION

TUBE, CAPILLARY, BLOOD SAMPLE

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-T-36487 (DSA-DM). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plain and heparin-coated capillary tubes for use in microhematocrit procedures.

1.2 Classification.

1.2.1 Types. Capillary tubes shall be of the following types, as specified for the appropriate stock number and item identification (see 6.2):

Type I - Heparinized
Type II - Plain

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-Up.

FSC 6630

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Glass shall be soda-lime glass, suitable for the intended purpose.

3.1.2 Heparin. Heparin shall be the ammonium salt isolated from lungs or the intestinal mucosa of beef or pork origin.

3.2 Construction. Capillary tubes shall be fabricated of glass, and shall be straight and open at both ends without lip or constriction. Bore of the capillary tube shall be uniform, and taper shall not exceed 0.001 inch (in.) in 3 in. Tube ends shall be cut at 90 degrees to tube axis, and shall not have sharp, chipped or jagged ends. The capillary tubes shall have the following dimensions (in millimeters):

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Inside diameter.....1.07 to 1.24
Length.....75 ± 2
Wall thickness.....0.20 +.03 -.02

3.2.1 Type I. Inner surfaces of type I tube shall be completely coated with ammonium heparin. Tube shall contain a minimum of 3.0 U.S.P. units, when tested as specified in 4.4.1. Before the coating process is applied, the tubes shall be clean and dry, and free of foreign material.

3.2.2 Type II. Type II tubes shall be plain and free of foreign material. Tubes shall be clean and dry, and ready for use.

* 3.3 Color coding. Each capillary tube shall be color coded. The type I tube shall have a red tip and the type II tube shall have a blue tip.

3.4 Performance. The inner surfaces, except as specified, shall be clean and free of filmy material which may impair the performance (see 3.4.1 and 3.4.2) of the tube.

3.4.1 Capillarity. The capillary tube shall be capable of drawing up blood or sheep plasma to a level of not less than 2 centimeters (cm.) from the top end when tested as specified in 4.4.2.

3.4.2 Fluidity (For type I only). No coagulation of the blood or sheep plasma shall be evident when tested as specified in 4.4.3.

* 3.5 Workmanship. Workmanship shall be first class throughout. The capillary tubes shall be free from defects which detract from their appearance or may impair their serviceability. The bore of the capillary tube shall be free of constrictions that interfere with drawing up blood.

* 3.6 Lot or control number (For type I only). A lot or control number shall be indicated on the interior packages and exterior containers (see 5.3) which can be traceable to specific production run(s) involving the same batch of ammonium heparin.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible

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for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit of product for inspection purposes shall be the individual capillary tube. No more than five capillary tubes shall be selected from any one unit package (see 5.1.1). For count, the unit of product for sampling purposes shall be the unit package.

Table I For examination

	Inspection level	AQL percent defective
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5
For taper examination	S-3	1.0
For count	S-1	*

* Acceptance number shall be zero.

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The acceptance number shall be zero. The unit of product for testing of capillary tubes shall be the unit package. The fluidity test shall be run on each lot of ammonium heparin material used for the type I capillary tubes.

Table II For tests

Characteristic	Requirement	Test procedure	Inspection level
Potency	3.2.1	4.4.1	S-4 *
Capillarity	3.4.1	4.4.2	S-1 **
Fluidity	3.4.2	4.4.3	S-1 **

* For purposes of this test, 10 packages shall be combined into one test sample. If the number of packages from the sampling plan is not divisible by 10, a sufficient number of packages shall be added to the odd packages to make a test sample.

** One tube shall be selected from a unit package. No more than one unit package may be selected from an intermediate package (see 5.1.2).

4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Table III Categories and defects *

Major	
101	Tube not free of cracks or breaks.
102	Tube not free of sharp, chipped, or jagged ends.
103	Inside surface not completely coated (type I only).
104	Tube not clean or dry, or free of foreign material.
105	Tube not cut at 90 degrees to the axis of the tube.
106	Tube not straight.
Minor	
201	Color code on tube not complete.
202	Glass not free of scratch marks.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Taper examination. The tubes shall be examined for defects in taper of the bore. Any taper not within the tolerance specified herein shall be classified as a defect.

4.3.4 Count. The number of tubes in the unit package shall be determined and shall be not less than 95 tubes for an individual package; the average count shall be not less than 100 tubes per unit package. The number of tubes in the unit package shall be determined by weighing 100 tubes, and calculating the number per sample package based on weighing the total quantity in a sample package.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Heparin content. Heparin content shall be determined by the method of assaying sodium heparin in the latest edition of the U.S.P. The results obtained shall represent the heparin content on the inner surfaces of the tubes. No heparin from the outside surface shall enter the test sample.

4.4.2 Capillarity test. Sample capillary tubes shall be tested in accordance with currently accepted technique for hematocrit determination and the level to which the blood is drawn up, within a 30-second period, shall be noted. Sheep plasma may be used in lieu of blood and shall be prepared in accordance with U.S.P. Assay for Sodium Heparin.

4.4.3 Fluidity test. The test shall be conducted by initially preparing recalcified sheep plasma by the following two step process:

a. Sheep plasma preparation. Sheep plasma shall be prepared in accordance with U.S.P. Assay for Sodium Heparin.

b. Recalcify sheep plasma. To 10 milliliters (ml.) of the prepared sheep plasma shall be added 2.0 ml. of the 1.0% calcium chloride solution used in the heparin assay (4.4.1). The sheep plasma and calcium chloride solution shall be mixed well.

In addition, samples of both the plain sheep plasma and recalcified sheep plasma shall be used as controls in accordance with the following:

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a. Positive control. A plain (i.e. non-heparinized) capillary tube shall be filled with recalcified sheep plasma.

b. Negative control. A heparinized capillary tube shall be filled with plain sheep plasma (see 4.4.3 a).

Immediately after the preparation of the recalcified sheep plasma, the capillary tubes shall be filled by immersing the tips in the recalcified sheep plasma and holding the tubes at such an angle as to facilitate quick filling. The capillary tubes shall be filled to 3 mm. from the end. At the end of a one-hour period, the sample tubes shall be inspected for evidence of coagulation by carefully snapping off segments of tubing which shall be placed flat on a surface of black background to facilitate observation and comparing with the control samples. Evidence of coagulation occurring is indicated by the sheep plasma becoming opaque or the presence of a fine fibrin thread.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package, containing 100 tubes as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a plastic or paperboard vial of suitable size and design. Capillary tubes shall be suitably packaged to prevent breakage.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676 or PPP-B-636, type CF, class domestic, except that commercial colors shall be acceptable. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Fiberboard boxes shall conform to the special requirements specified in PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Fiberboard boxes shall conform to special requirements specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

* 5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. The lot (control) number shall also be shown for type I capillary tubes.

* 5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. The lot (control) number shall also be shown for type I capillary tubes.

6. NOTES:

6.1 Intended use. Capillary tubes shall be suitable for use with:

a. FSN 6640-299-9835 Centrifuge, Laboratory, Microhematocrit, 110 Volt, 60 cycle

b. FSN 6630-585-1378 Reader, Microhematocrit, Spiral type.

* 6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Type of tube required (see 1.2.1).
d. Federal Stock Number (FSN).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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